

11. Usando la propiedad $P(A \cup B) = P(A) + P(B) - P(A \cap B)$, probar la fórmula

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

$$P(A \cup B \cup C) = P([A \cup B] \cup C)$$

$$= P(A \cup B) + P(C) - P[(A \cup B) \cap C]$$

$$= P(A \cup B) + P(C) - P[(A \cap C) \cup (B \cap C)]$$

$$= P(A) + P(B) - P(A \cap B) + P(C) - [P(A \cap C) + P(B \cap C) - P[(A \cap C) \cap (B \cap C)]]$$

$A \cap B \cap C$

$$P(A \cup B \cup C) = P(A) + P(B) - P(A \cap B) + P(C) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$